

CSL301 : Operating System

Quiz-1

Marks : 100

Time : 40 Min

1. Suppose the following jobs are to be executed in a uniprocessor system. Multilevel Feedback Queue(MLQ) is used with queues numbered 1-10, quantum = $2i$, where i is the queue level number and processes are initially placed in the first queue (i.e., level 1). In this scheduling policy, each process executes at a particular level for one quantum and then moves down a level; processes never move up a level and assume context switch delay of 1ms. Calculate the average process turnaround time, average waiting time, average response time and make a Gantt chart. **40 Marks**

Process	Arrival time	Burst
1	0	4
2	1	8
3	3	2
4	10	6
5	12	5

Probable Solution:

- Uniprocessor system
- Context switch time = 1 (assumed)
- Quantum = $2i$

Queue Level	Quantum
Q1	$2(1)=2(1)=2$ ms
Q2	$2(2)=4(2)=4$ ms
Q3	$2(3)=6(3)=6$ ms

Q4	$2(4)=82(4)=8$ ms
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	0	2	3	5	6	8	9	11	12	14	15	17	18	22	23	27	28	31	32	34
process	P ₁	CS	P ₂	CS	P ₃	CS	P ₁	CS	P ₄	CS	P ₅	CS	P ₂	CS	P ₄	CS	P ₅	CS	P ₂	
queue	Q ₁		Q ₂		Q ₁		Q ₂		Q ₁		Q ₁		Q ₂		Q ₂		Q ₂		Q ₃	

Process	Arrival (AT)	Burst (BT)	First CPU Allocation (FCA)	CT	TAT (CT - AT)	WT (TAT - BT)	RT (FCA - AT)
P1	0	4	0	11	11	7	0
P2	1	8	3	34	33	25	2
P3	3	2	6	8	5	3	3
P4	10	6	12	27	17	11	2
P5	12	5	15	31	19	14	3
Average					17	12	2

- P1: P1 first gets the CPU in Queue 1 (Q1) for 2 time units. Since its total burst time is 4, it has 2 time units remaining. Therefore, P1 moves to Queue 2 (Q2).

- P2: P2 then gets the CPU in Q1 for 2 time units. Its total burst time is 8, so 6 time units remain. Therefore, P2 moves to Q2.
- P3: P3 gets the CPU in Q1 for 2 time units. Since its required burst time is exactly 2 time units, P3 completes its execution.
- P1: P1 now gets the CPU in Q2. The time quantum of Q2 is 4 time units, but P1 needs only 2 more time units. Therefore, P1 completes its execution.
- P4: P4 gets the CPU in Q1 for 2 time units. Since its burst time is 6, 4 time units remain. Therefore, P4 moves to Q2.
- P5: P5 gets the CPU in Q1 for 2 time units. Since its burst time is 5, 3 time units remain. Therefore, P5 moves to Q2.
- P2: P2 now gets the CPU in Q2. The Q2 quantum is 4 time units. P2 uses all 4 time units, leaving 2 time units remaining. Therefore, P2 moves to Q3.
- P4: P4 gets the CPU in Q2 for 4 time units. These 4 units are exactly what P4 needs to finish, so P4 completes its execution.
- P5: P5 gets the CPU in Q2. Although the Q2 quantum is 4 time units, P5 needs only 3 more time units, so P5 completes its execution.
- P2: Finally, P2 gets the CPU in Q3. The Q3 quantum is $2 \times 3 = 6$ time units, but P2 requires only 2 more time units, so P2 completes.

2. Consider the 3 processes, P1, P2 and P3 shown in the table.

Process	Arrival time	Time Units Required
P1	0	5
P2	1	7
P3	3	4

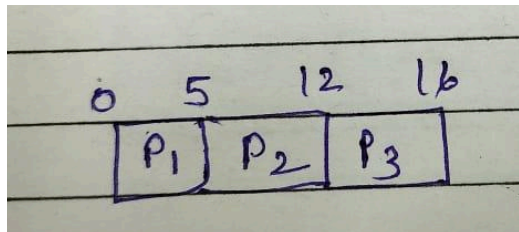
Calculate the average process turnaround time, average waiting time, average response time and make Gantt charts under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units).

30 Marks

Probable Solution:

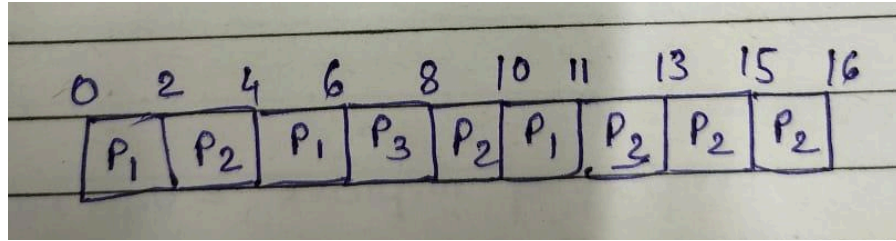
a. FCFS Scheduling

Process	Arrival (AT)	Burst (BT)	First CPU Allocation (FCA)	CT	TAT (CT - AT)	WT (TAT - BT)	RT (FCA - AT)
P1	0	5	0	5	5	0	0
P2	1	7	5	12	11	4	4
P3	3	4	12	16	13	9	9
Average					9.67	4.33	4.33



b. Round Robin RR2:

Processes	Arrival (AT)	Burst (BT)	First CPU Allocation (FCA)	CT	TAT (CT - AT)	WT (TAT - BT)	RT (FCA - AT)
P1	0	5	0	11	11	6	0
P2	1	7	2	16	15	8	1
P3	3	4	6	13	10	6	3
Average					12.00	6.67	1.33



3. An operating system uses Shortest Remaining Time first (SRT) process scheduling algorithm. Consider the arrival times and execution times for the following processes:

Process	Execution time	Arrival time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

Calculate the average process turnaround time, average waiting time, average response time and make a Gantt chart.

Probable Solution:

Processes	Arrival (AT)	Burst (BT)	First CPU Allocation (FCA)	CT	TAT (CT - AT)	WT (TAT - BT)	RT (FCA - AT)
P1	0	20	0	20	20	0	0
P2	15	25	20	55	40	15	5
P3	30	10	30	40	10	0	0
P4	45	15	55	70	25	10	10
Average					23.75	6.25	3.75

0	20	30	40	55	70
P_1	P_2	P_3	P_2	P_4	